

Euler's Method and Logistic Growth

ANSWER KEY



The Euler's Method formula

- 1 State the recursive formula used in Euler's Method to approximate solutions to $\frac{dy}{dx} = f(x, y)$.
 $y_{n+1} = y_n + f(x_n, y_n) \cdot \Delta x$, starting from the given initial point (x_0, y_0) and stepping forward by Δx each iteration.
- 2 Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$.
Step 1: $x_0 = 0, y_0 = 1$. $y_1 = 1 + (0 + 1)(0.5) = 1.5$ at $x_1 = 0.5$. Step 2:
 $y_2 = 1.5 + (0.5 + 1.5)(0.5) = 1.5 + 1 = 2.5$ at $x_2 = 1$. Estimate: $y(1) \approx 2.5$.
- 3 Given $\frac{dy}{dx} = y$ with $y(0) = 1$, use Euler's Method with $\Delta x = 0.5$ to estimate $y(1)$, then compare to the exact solution $y = e^x$.
Step 1: $y_1 = 1 + (1)(0.5) = 1.5$ at $x = 0.5$. Step 2: $y_2 = 1.5 + (1.5)(0.5) = 2.25$ at $x = 1$. Estimate: $y(1) \approx 2.25$. Exact: $y(1) = e \approx 2.718$. Euler's Method underestimates.

Euler's Method with smaller step sizes

- 4 Given $\frac{dy}{dx} = 2x$ with $y(0) = 0$, use Euler's Method with $\Delta x = 0.25$ to estimate $y(0.5)$, then compare to the exact solution $y = x^2$.
Step 1: $x_0 = 0, y_0 = 0$: $y_1 = 0 + 2(0)(0.25) = 0$ at $x = 0.25$. Step 2: $y_2 = 0 + 2(0.25)(0.25) = 0.125$ at $x = 0.5$. Estimate: $y(0.5) \approx 0.125$. Exact: $y(0.5) = 0.25$. Euler's Method underestimates here.
- 5 For $\frac{dy}{dx} = 2x$ with $y(0) = 0$, redo the estimate for $y(0.5)$ using $\Delta x = 0.1$ (5 steps), and compare accuracy to the $\Delta x = 0.25$ result.
Steps (x : 0, 0.1, 0.2, 0.3, 0.4; y : 0, 0, 0.02, 0.06, 0.12): at each step $y_{n+1} = y_n + 2x_n(0.1)$. Final: $y(0.5) \approx 0.12 + 2(0.4)(0.1) = 0.12 + 0.08 = 0.20$. This is closer to the exact value 0.25 than the $\Delta x = 0.25$ estimate (0.125) — smaller step size gives better accuracy.

Error analysis in Euler's Method

- 6 Explain why Euler's Method tends to underestimate y when the solution curve is concave up, using a tangent-line reasoning argument.
Euler's Method uses the tangent line at each step to predict the next point. When the curve is concave up, the actual curve lies above its tangent lines, so following the tangent line repeatedly causes the approximation to fall below the true curve — an underestimate.
- 7 Explain what happens to the accuracy of Euler's Method as the step size $\Delta x \rightarrow 0$.
As $\Delta x \rightarrow 0$, each linear approximation step spans a shorter interval, so the tangent-line error accumulated per step shrinks, and the overall approximation converges to the true solution curve.

Logistic growth models

- 8 State the logistic differential equation and identify the carrying capacity L in $\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right)$.
General form: $\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$, where k is the growth rate and L is the carrying capacity. Here $k = 0.3$ and $L = 500$.
- 9 For the logistic model $\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right)$, find the population P at which growth rate is maximized.
The growth rate $\frac{dP}{dt}$ is maximized when $P = \frac{L}{2} = \frac{500}{2} = 250$ (the inflection point of the logistic curve).
- 10 The logistic model with $L = 500$, $k = 0.3$, and $P(0) = 50$ has closed-form solution $P(t) = \frac{500}{1 + 9e^{-0.3t}}$. Sketch its shape and identify the horizontal asymptote.
As $t \rightarrow \infty$, $e^{-0.3t} \rightarrow 0$, so $P(t) \rightarrow \frac{500}{1} = 500$. The horizontal asymptote is $P = 500$ (the carrying capacity), and the curve is S-shaped (sigmoid), starting near 50, growing fastest near $P = 250$, then leveling off toward 500.
- 11 For a logistic model with carrying capacity $L = 1000$ and $k = 0.2$, find $\frac{dP}{dt}$ when $P = 200$ and when $P = 800$, and explain which is closer to the maximum growth rate.
 $\frac{dP}{dt} = 0.2P\left(1 - \frac{P}{1000}\right)$. At $P = 200$: $0.2(200)(0.8) = 32$. At $P = 800$: $0.2(800)(0.2) = 32$ (equal, by symmetry about $P = 500$). Neither is at the true maximum ($P = 500$, where $\frac{dP}{dt} = 0.2(500)(0.5) = 50$); both are symmetric points with equal, smaller growth rates.

Interpreting logistic growth graphically

- 12 Explain what happens to $\frac{dP}{dt}$ as $P \rightarrow L$ in the logistic model, and what this means for the population over time.
As $P \rightarrow L$, the factor $\left(1 - \frac{P}{L}\right) \rightarrow 0$, so $\frac{dP}{dt} \rightarrow 0$ — growth slows and the population levels off, approaching the carrying capacity L asymptotically rather than exceeding it.
- 13 Explain why the logistic model is considered more realistic than the unrestricted exponential model $\frac{dP}{dt} = kP$ for real populations.
The exponential model predicts unbounded growth forever, which is unrealistic since resources (food, space) are finite. The logistic model incorporates a carrying capacity L that naturally slows growth as the population approaches the environment's limit, matching real-world population behavior.

Building an Euler's Method table

- 14 Given $\frac{dy}{dx} = x - y$ with $y(0) = 2$, complete an Euler's Method table with $\Delta x = 0.5$ for $x = 0, 0.5, 1, 1.5, 2$, showing each y_n .
 $x_0 = 0, y_0 = 2$. $y_1 = 2 + (0 - 2)(0.5) = 1$ at $x = 0.5$. $y_2 = 1 + (0.5 - 1)(0.5) = 0.75$ at $x = 1$.
 $y_3 = 0.75 + (1 - 0.75)(0.5) = 0.875$ at $x = 1.5$. $y_4 = 0.875 + (1.5 - 0.875)(0.5) = 1.1875$ at $x = 2$.
Table: $(0, 2), (0.5, 1), (1, 0.75), (1.5, 0.875), (2, 1.1875)$.

- 15 Using the table above, estimate $y(2)$ and state whether the true solution is likely above or below this estimate near $x = 2$, given the slope $\frac{dy}{dx} = x - y$ becomes positive once $y < x$.
 Estimate: $y(2) \approx 1.1875$. Since $y_4 = 1.1875 < x_4 = 2$, the slope is positive and the curve is rising there; without more precise second-derivative information the direction of Euler's error can't be determined from this alone, but the table shows the solution turning around near $x = 1$ where y stops decreasing and begins increasing.

Differential equations as models of real phenomena

- 16 A cup of coffee cools according to Newton's Law of Cooling, $\frac{dT}{dt} = -0.15(T - 20)$, with $T(0) = 90^\circ\text{C}$. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.
 $T_0 = 90$. $T_1 = 90 + (-0.15)(90 - 20)(1) = 90 - 10.5 = 79.5$.
 $T_2 = 79.5 + (-0.15)(79.5 - 20)(1) = 79.5 - 8.925 = 70.575$. Estimate: $T(2) \approx 70.6^\circ\text{C}$.
- 17 Explain why Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_{\text{room}})$, has the same mathematical structure as exponential decay shifted by a constant, and state its equilibrium (constant) solution.
 Substituting $y = T - T_{\text{room}}$ gives $\frac{dy}{dt} = -ky$, the standard exponential decay equation. So $T(t) = T_{\text{room}} + (T_0 - T_{\text{room}})e^{-kt}$. The equilibrium solution is $T = T_{\text{room}}$ (where $\frac{dT}{dt} = 0$), the temperature the object approaches as $t \rightarrow \infty$.

Comparing Euler's Method to the exact logistic solution

- 18 For $\frac{dP}{dt} = 0.3P(1 - \frac{P}{500})$ with $P(0) = 50$, use Euler's Method with $\Delta t = 2$ to estimate $P(2)$, then compare to the exact value $P(2) = \frac{500}{1 + 9e^{-0.6}} \approx 121.9$ (using the closed-form solution from earlier).
 $P_0 = 50$. At $P = 50$: $\frac{dP}{dt} = 0.3(50)(1 - \frac{50}{500}) = 0.3(50)(0.9) = 13.5$. $P_1 = 50 + 13.5(2) = 77$. Euler's estimate $P(2) \approx 77$ is noticeably below the exact value ≈ 121.9 — a large step size ($\Delta t = 2$) on a fast-growing region of the logistic curve produces significant underestimation.

Autonomous differential equations and equilibria

- 19 A differential equation is called autonomous if $\frac{dy}{dx}$ depends only on y , not on x . Explain why both $\frac{dy}{dx} = ky$ and the logistic equation are autonomous, and find the equilibrium solutions of $\frac{dy}{dx} = y(y - 4)$.
 Both equations have right-hand sides depending only on y (ky and $kP(1 - P/L)$), with no explicit x or t . Equilibria occur where $\frac{dy}{dx} = 0$: $y(y - 4) = 0 \Rightarrow y = 0$ or $y = 4$.
- 20 For $\frac{dy}{dx} = y(y - 4)$, determine whether the equilibrium $y = 0$ is stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below $y = 0$.
 Just below $y = 0$ (e.g. $y = -1$): $(-1)(-5) = 5 > 0$, so y increases toward 0. Just above $y = 0$ (e.g. $y = 1$): $(1)(-3) = -3 < 0$, so y decreases toward 0. Since solutions move toward $y = 0$ from both sides, it is a STABLE equilibrium.